ADULT CARDIAC SURGERY

1. Prevent & be prepared

- Ensure adequate training of the staff in technical skills and ALS
- Ensure availability and functioning of emergency equipment
- Use safety checklists
- Identify and manage deterioration in the post-operative cardiac patient

2. Detect cardiac arrest and activate cardiac arrest protocol

- Confirm cardiac arrest by clinical signs and pulseless waveforms
- Call for help and activate cardiac arrest protocol
- • Ensure airway patency and breathing
  • Deliver 100% oxygen
  • Stop syringe drivers
- Consider echocardiography to early identify possible reversible causes

3. Resuscitate and treat possible reversible causes

ASSESS RHYTHM

VF/pVT	Asystole / extreme bradycardia	PEA
Defibrillate (apply up to 3 consecutive shocks)	Pace - if wires available	Provide high-quality CPR

Start basic life support and address reversible causes

Amiodarone 300 mg IV	Consider external pacing if wires not available	If paced, turn off pacing to exclude underlying VF

No ROSC
- Continue advanced life support
- Prepare & perform early resternotomy (<5 min)
- Consider circulatory support devices and ECPR

GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

ADULT CARDIAC CATHETERISATION LABORATORY

1. Prevent & be prepared

- Promote adequate training of the staff in technical skills & ALS
- Ensure emergency equipment is functional and readily available
- Plan elective procedures carefully and use safety checklists

2. Detect and react

- Check patient's status and monitored vital signs regulary
- Consider echocardiography if haemodynamic instability or complication
- In case of cardiac arrest, call for help & activate the resuscitation team

3. Resuscitate and treat possible causes

VF / pVT cardiac arrest
→ **Defibrillate** (apply up to 3 consecutive shocks)
→ **No ROSC**

Asystole / PEA
→ Consider **external or transvenous pacing** in extreme bradycardia
→ **No ROSC**

Continue CPR according to the ALS algorithm

- Check & correct reversible causes, including **ultrasound & angiography**
- Consider **mechanical CPR** to facilitate chest compression
- Consider **ECPR** in refractory cardiac arrest
- Consider **circulatory support devices** in case of shock after ROSC

ADULT CORONARY THROMBOSIS

SPECIAL CIRCUMSTANCES

1. Prevent & be prepared

- Promote cardiovascular health to reduce the risk of acute coronary events
- Enhance health education to recognise symptoms & BLS training for likely rescuers
- Strengthen regional STEMI networks to ensure timely PCI

2. Suspect & react

- Evaluate 12-lead ECG after ROSC; repeat if incon-clusive
- Assess clinical data suggesting acute coronary syndrome
- Activate the STEMI network & transfer the patient to PCI centre if ST-elevation or ongoing ischemia

3. Resuscitate & establish reperfusion strategy

Patients with sustained ROSC

ST-elevation	No ST-elevation
Is PCI possible within 120 min?	**Is the patient unstable or shows signs of ongoing ischemia?**
Yes Immediate coronary angiography ± PCI	**Yes** Immediate coronary angiography ± PCI
No Consider fibrinolysis if not contraindicated & transfer to PCI centre	**No** Consider ruling-out non-coronary causes Consider delayed coronary angiography ± PCI

No Sustained ROSC

Does the OHCA setting suggest likely futility?

Yes Consider stopping CPR

No Consider transfer to PCI centre with ongoing CPR
→ Consider mechanical CPR
→ Consider ECPR

SCAN FOR MORE HELPFUL MATERIALS

MODIFICATIONS TO ADVANCED LIFE SUPPORT IN OBSTETRIC PATIENTS

ALS DURING PREGNANCY	RATIONALE
Call for help – "maternal arrest team" (incl. an obstetrician and neonatologist)	Consider pregnancy in any collapsed woman of childbearing age.
Maintain manual (left) uterine displacement throughout resuscitation	Relieve aortocaval compression early to improve cardiac output. One performs manual left uterine displacement, the others CPR
Airway management	Aspiration and failed intubation risk are both increased. Use stepwise rescue (bag-mask, supraglottic airway if intubation fails). Use a ramped position. The most experienced should perform airway management, and consider: ▪ Short-handled laryngoscope (large breasts) ▪ Video laryngoscope = standard ▪ Smaller tracheal tube with guidance
Defibrillation pad position	Ensure pads are placed beneath breast, not over
Fetal monitoring	Remove internal & external fetal monitors before defibrillation
Identify common and reversible causes	4Hs, 4Ts and 4Ps
Early vascular access	IV/IO access above the diaphragm
Calcium chloride	10 ml IV calcium chloride 10 % - for Mg overdose, low calcium or hyperkalaemia
Magnesium	2 g IV: for polymorphic VT 4 g IV: for eclampsia
Tranexamic acid	1 g IV: for haemorrhage
Resuscitative Hysterotomy (Perimortem Caesarean section)	In patients >20 weeks' gestation or fundus above umbilicus. It is time-sensitive, early preparation, and immediate intervention at the cardiac arrest site by a skilled team.
Post-resuscitation care	Stabilise mother - stabilises fetus. Prepare for major obstetric haemorrhage.

ADULT TRAUMATIC CARDIAC ARREST ALGORITHM

